

Electronics as a Hobby



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Electronics is a unique and multifaceted discipline of modern science. It is looked at from different perspectives. Scholars and researchers critically study the theory of it, teachers inculcate it as an engineering knowledge, students learn it as a subject of their academic curriculum, manufacturers consider it a profitable commercial endeavour, retailers benefit from its sale, technicians practise it as a skilled trade and hobbyists regard it as a rewarding avocation.

Electronics as a hobby is a very popular leisure pursuit worldwide. It is a prolific source of entertainment and knowledge. Though highly technical, electronics as a

hobby can be pursued by all those who have working knowledge of the functioning of various components and reading of schematics. Perhaps this is why enthusiasts from diverse fields of activities are seen pursuing this hobby. For many, electronics as a hobby is a lifelong companion. Contrary to the electrical circuits that work on dangerous mains voltage, basic electronics circuits generally work on low DC voltages and hence are not dangerous to experiment with.

In addition to this, affordable prices of components due to their mass production, easy availability of components on counters and through online purchases and a lot of information easily accessible on the Internet have together contributed to the popularity of this hobby. Many hobby circuits are not just entertaining but have wide applications, too.

Occasionally, a hobbyist while fiddling with a circuit is likely to get surprised when the modified circuit actually functions better than expected. Even accidentally a totally new application is likely to crop up. For the hobbyist every successfully working circuit is a

creation and observing its functioning is a recreation.

Advice for beginners

Notwithstanding the rewards the hobby offers, it should be borne in mind that intermittent failures are part and parcel of this hobby. Nonetheless, the hobbyist must learn to win over failures without feeling defeated. Patience and perseverance are prerequisites to this hobby. Over-eagerness and haste are but an invitation to failures.

If prior to an experiment the connections are properly checked and the tips are properly followed, the rate of failure can be drastically reduced and a greater pleasure and satisfaction can be drawn. Juniors and beginners must not attempt circuits for which power supply is drawn from the mains. Carelessness and insufficient knowledge are fatal pitfalls.

To begin with, the circuit to be assembled should be studied properly. Beginners should not attempt apparently appealing radio frequency (RF) circuits like radios and transmitters in the budding stage of the hobby. These projects require a high degree of skill and practice. Instead, they should try basic projects using LED bulbs, general-purpose transistor projects and simple audio projects that need just a few components.

Once well-versed with the use of various components, they can try simple integrated circuit (IC) projects beginning with IC NE555, a simple but highly-versatile timer IC. Many simple and useful projects can be built using NE555. Slowly and steadily hobbyists must thread their way through new experiences and enter the domain of electronics to find an expanse of entertainment and knowledge waiting for them.

How to get started

When a particular circuit is decided to be assembled, required components shown in the schematic should be kept ready. Always stick to the schematic. Only the specified components, particularly semiconductors,

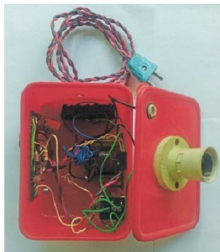


Fig. 1: Clap lamp boxed up in a red plastic box; in working order since about three decades

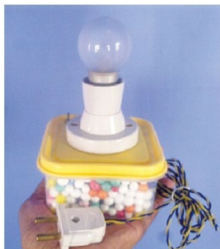


Fig. 2: Touch-lamp built using a field-effect transistor